

Manufacturing & Production Playbook

How to design, source, produce, decorate, and ship your product without getting scammed or drowning in chaos.

Who this is for: Fashion brand owners building real products (not IG moodboards).

What you get: Production models, step-by-step workflow, key terms, fabric choices by product/season, and a practical guide to printing & decoration (DTF/DTG/screen/sublimation/embroidery and more).

How to use: Read once. Then use the checklists every time you plan a drop.

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1) The Manufacturing Map

Manufacturing is not 'find a factory'. It is a chain: design intent → specs → sampling → costing → bulk production → quality control → finishing → packing → delivery. Every break in this chain costs you money and reputation.

Golden rule: You do not control quality by 'telling the factory'. You control quality by defining measurable standards (specs, tolerances, tests) and enforcing them at checkpoints.

Checkpoints you must own:

- Tech pack + BOM ready (no missing details).
- Sample approval (fit, construction, fabric, print/embroidery).
- PP sample (pre-production) approved.
- Inline QC during sewing + decoration.
- Final QC before packing/shipping.

2) Production Models

Different production models exist for different risk levels. If you pick the wrong model, you will either overstock, run out of stock, or destroy margin with inefficient setup costs.

Model	What it is	Pros	Cons	Best for
Made-to-Order (MTO)	Produce after a customer pays.	Low inventory risk, cash-in first.	Longer delivery, harder scaling.	Custom pieces, premium drops, early-stage proof.
Small Batch	Low quantity runs (usually near MOQ).	Tests demand with limited risk.	Higher unit cost.	New styles, new sizing, uncertain demand.
Batch Production	Standard bulk runs on a schedule.	Lower unit cost, stable supply.	Inventory risk.	Proven best-sellers, repeat styles.
Flow / Line Production	Continuous output optimized for speed.	Fast, low cost at scale.	Requires stable demand and processes.	High-volume basics (tees, socks, underwear).
Print-on-Demand (POD)	Blank + printing per order (DTG/DTF).	No inventory, fast testing.	Lower margin, quality variability.	Early stage, content-driven brands, quick launches.

Ruthless mentor note: If you have no data, you should not be doing big batch. Batch production is not 'confidence' - it is a bet. Earn the right to bet with small batch or POD first.

3) When to Use Each Model

Use these rules to decide your production model. Stop guessing.

Situation	Default choice
You have no sales history	POD or Small Batch
You have repeat demand + stable sizing	Batch Production
You sell premium/custom and can wait	Made-to-Order
You sell high-volume basics weekly	Flow / Line Production
You need to test 3-5 designs quickly	POD or Micro-batch + DTF

Demand test ladder (use in order):

- Sketch + mockups (validate concept with audience).
- Prototype sample (validate look + fit).
- Micro-batch (validate sell-through + returns).
- Repeat batch (validate operational repeatability).
- Scale with line production only after stability.

4) Terminology You Must Know

Term	Meaning (in plain language)
SKU	Stock Keeping Unit. Unique code per product + color + size. Without SKUs you cannot manage inventory.
MOQ	Minimum Order Quantity. The lowest quantity a supplier accepts per color/style/fabric.
MVP	Minimum Viable Product. Smallest version that proves demand (not the cheapest version).
BOM	Bill of Materials. List of all fabrics, trims, thread, labels, packaging, and quantities.
Tech Pack	The factory blueprint: measurements, construction, stitch types, artwork placement, labels, etc.
GSM	Grams per square meter. Fabric weight - critical for feel and season.
PP Sample	Pre-production sample. Final checkpoint before bulk starts.
Strike-off	Print sample test (color/hand feel) before bulk printing.
Size Set	Samples in all sizes to confirm grading (S/M/L etc).
QC	Quality Control. Must be defined with measurable tolerances.
AQL	Acceptable Quality Limit. Sampling standard for defect acceptance.
Lead Time	Time from PO to delivery. Includes fabric sourcing, production, and shipping.

5) From Idea to Bulk: End-to-End Workflow

Stage	What you do
1. Define product spec	Category, fit, target price, season, fabric weight, decoration method.
2. Build tech pack	Measurements, construction, BOM, labels, artwork placement, tolerances.
3. Find supplier shortlist	3 suppliers minimum. Compare capability, MOQ, lead time, sample cost.
4. Sampling	Prototype → revisions → fit approval → print/embroidery test.
5. Costing & margin check	Use your costing tool. If margin is trash, redesign before bulk.
6. PP sample	Final sign-off. Lock specs. No last-minute changes.
7. Bulk production + inline QC	Check during sewing and decoration. Do not wait until the end.
8. Final QC + packing	Measure, test print wash, check labels, packaging, barcodes.
9. Shipping + receiving	Count cartons, check random units, log defects and supplier issues.

Non-negotiable: Lock your specs at PP sample. If you keep changing designs mid-bulk, you are paying for chaos.

6) Suppliers & Sourcing

Factories are not equal. Choose based on capability (product type + decoration), not price alone. Cheap suppliers are expensive once you count rework, delays, and refunds.

Supplier evaluation checklist:

- Do they already make your product category (ask for similar references)?
- Can they handle your decoration method in-house (print/embroidery) or do they outsource?
- Do they provide consistent fabric sourcing (same mill or stable alternatives)?
- Do they accept measurable QC standards (tolerances, AQL) or do they argue emotionally?
- Do they provide production schedule + weekly updates?

Red flag: A supplier that refuses to put details in writing is a supplier planning to 'forget' later.

7) Fabrics & Materials by Product + Season

Fabric choice makes or breaks quality. Pick fabric based on: season, drape, durability, shrinkage, colorfastness, and how it behaves with printing/embroidery.

Best default fabrics by product:

Product	Best default fabrics	Notes (quality & season)
T-shirt (streetwear)	100% cotton jersey 180-240 GSM (combed ring spun)	Heavier for premium feel. Pre-shrunk preferred.
Hoodie	Cotton fleece 320-450 GSM (brushed back)	Higher GSM = warmer + premium. Watch shrinkage.
Sweatpants	Fleece 280-380 GSM or French terry	French terry = breathable. Brushed fleece = winter.
Shirt (woven)	Cotton poplin, oxford, or twill	Poplin for crisp SS. Twill/oxford more durable.
Denim	Cotton denim 10-14 oz	Heavier for jackets; midweight for jeans.
Activewear	Polyester/spandex blends (4-way stretch)	Avoid heavy prints that crack on stretch.
Dresses	Viscose/rayon, cotton voile, satin, chiffon	Choose based on drape. Test shrink and color bleed.

Season guidance (simple):

- Spring/Summer: lower GSM, breathable weaves (jersey 160-200, poplin/voile, linen blends).
- Autumn/Winter: higher GSM, brushed fleece, heavier twills, lined outerwear fabrics.
- Transitional (Egypt reality): midweight is king; avoid extreme winter-only fabrics unless your audience wants it.

8) Trims, Hardware, and Construction

Trims are where cheap brands get exposed: labels peeling, zippers breaking, drawcord tips falling off. Specify trim quality in BOM and test it.

Key trim decisions:

- Labels: woven vs printed vs heat transfer. Woven feels premium but costs more.
- Thread: match thread type to fabric and stress points (poly core for strength).
- Stitching: single needle, coverstitch, overlock - define where each is used.
- Hardware: zippers (YKK or equivalent), snaps, rivets - define finish and supplier.
- Reinforcement: bartacks at pockets, stress points, zipper ends.

Durability hack: Add bartacks and double-stitching to stress points. It costs little and reduces returns.

9) Costing, Negotiations, and Payment

If you don't cost properly, you will price emotionally. Use your calculator and include everything: fabric, trims, printing, packaging, shipping, rejects, and returns.

Typical payment structures:

Structure	When used	Risk
50/50 (deposit/balance)	Most small factories	Medium. You still need QC checkpoints.
30/70 with PP approval	Better suppliers	Lower risk if PP sample is enforced.
LC / escrow	Large orders	Lower payment risk; higher admin.

Negotiation that actually works: Don't beg for a lower unit price. Ask for: better MOQs, faster lead time, free rework for defects, and stable fabric sourcing. Those win you money long-term.

10) Quality Control and Failures

Quality is not 'looks good'. It's measurable. Define tolerances, then check against them.

QC checklist:

- Measurements: specify tolerances (e.g., chest +/- 1 cm).
- Shrinkage: pre-wash test on fabric and sample.
- Color: check against Pantone or approved swatch.
- Seams: inspect stitch density, skipped stitches, broken thread.
- Print: wash test, stretch test (if needed), crack/peel test.

Reality: If you only do QC at the end, your only option is accepting trash or throwing money away. Inline QC gives you leverage while the factory is still working.

11) Production Planning (Lead Times & Risk)

Most delays are predictable: fabric sourcing, sampling loops, print approvals, and shipping. Plan backwards from launch date and add buffers.

Phase	Typical time (small factory)	Buffer you should add
Sampling loop	2-4 weeks	+1-2 weeks
Fabric sourcing	1-3 weeks	+1 week
Bulk sewing	2-4 weeks	+1 week
Printing/embroidery	3-10 days	+3 days
Packing + shipping	1-3 weeks	+1 week

Operational rule: If your launch depends on 'everything goes perfectly', you don't have a plan.

12) Printing & Decoration: Methods + Best Fit

Decoration is not just aesthetics - it changes cost, durability, hand feel, and even how the garment breathes. Pick a method based on fabric, design complexity, quantity, and price point.

Major methods you will actually use:

- Screen printing (plastisol, water-based, discharge)
- DTG (Direct-to-Garment)
- DTF (Direct-to-Film)
- Sublimation
- Heat transfer vinyl (HTV)
- Embroidery (flat, 3D puff, chenille)
- Applique/patches (woven/embroidered patches, chenille patches)

Ruthless mentor note: If you're doing 'premium streetwear' and your prints feel like plastic armor, your product is not premium. Hand feel matters.

12.1) Print Method Deep Dives

Method	Best for	Pros	Cons	Notes
Screen print - Plastisol	Tees, hoodies (bulk)	Durable, vibrant, great for high volume	Setup cost per color, heavier hand feel	Best when 50+ pcs per design per colorway.
Screen print - Water-based	Premium tees	Soft hand feel, breathable	Can be less vibrant on darks, needs good shop	Great for 'premium' feel. Needs underbase on darks.
Screen print - Discharge	Dark cotton tees	Very soft, vintage look	Not for synthetics, color results vary	Test on fabric lot. Not predictable like plastisol.
DTG	Complex full-color, low qty	Photo-quality, no setup per color	Slower, costlier per unit, best on cotton	Great for samples, small drops, ecommerce.
DTF	Most fabrics, small/medium qty	Works on cotton/poly blends, sharp detail	Hand feel can be heavier than DTG	The default 'flexible' method in many markets.
Sublimation	Polyester (sportswear)	No cracking/peeling, breathable	Only on light polyester, color shifts possible	Best for performance tees, leggings, jerseys.
HTV (vinyl)	Names/numbers, simple graphics	Fast, good for customization	Not great for big designs, can peel	Use for sports numbers, minimal logos.
Embroidery (flat)	Caps, hoodies, polos	Premium perception, durable	Not good for tiny details, can distort fabric	Specify stitch count/coverage. Use stabilizers.
3D Puff embroidery	Caps, hoodies	High-impact look	Bulkier, needs skilled operator	Use for bold logos, not detailed art.

12.2) Best Print/Decoration Choices by Product

Use this as a default. You can break the rules only if you know exactly why.

Product	Best default	Premium option	Avoid / watch out
100% cotton tee (streetwear)	Water-based screen OR DTG (small qty)	Discharge (on dark cotton) or high-quality screen with soft-hand ink	Cheap DTF with thick film; plastisol bricks on premium tees
Heavy tee 220-260 GSM	Screen print (soft-hand) or DTF (small batch)	Water-based + oversized placement with proper underbase	DTG on poor pre-treated fabric (fades fast)
Hoodie / fleece	Plastisol screen (bulk) or embroidery chest logo	Chenille patch / applique, mixed media	Huge DTF front prints with stiff feel
Sweatpants	Embroidery small logo, HTV for minimal	Applique patch	Large prints on high-stretch areas (cracking)
Polyester sports tee	Sublimation	Sublimation + embroidery patch (careful)	DTG (bad on poly), discharge (won't work)
Caps	Embroidery (flat or 3D puff)	Patch (woven/embroidered) + embroidery	DTF/DTG (inconsistent on curved panels)
Denim jacket	Embroidery patch + back screen print (with correct ink)	Applique + embroidery	DTG (poor), cheap transfers (peel)
Leather / faux leather	Emboss/engrave, patches	High-quality emboss + hardware	Heat methods without testing (melting risk)
Socks	Knit-in design (best)	Jacquard knit	Surface prints that crack after washes

Fast decision: If you are early-stage and need flexibility: DTF. If you are scaling a winning design: screen print. If it's polyester performance: sublimation.

12.3) Prints vs Fabrics vs Season

Print durability depends on fabric behavior. Stretch, heat, and wash cycles kill bad decoration choices.

- Stretch fabrics (spandex blends): prefer sublimation, stretch inks, or embroidery/patches. Avoid thick films.
- Hot climates: soft-hand prints (water-based) feel better. Heavy plastisol can feel sweaty.
- Winter fleece: embroidery and chenille feel premium and add texture without ruining hand feel.
- Dark garments: ensure underbase is correct or colors will look dead. Test every fabric lot.

13) Artwork Requirements (So Printers Don't Screw You)

Most print disasters come from bad artwork files. Send correct formats and approve strike-offs.

Method	What to send
Screen print	Vector (AI/EPS/PDF) preferred. Provide Pantone codes. Separate colors.
DTG/DTF	300 DPI PNG with transparent background. Provide print size in cm.
Sublimation	High-res artwork. Account for color shift. Use polyester-specific profiles if possible.
Embroidery	Vector preferred. Keep details bold. Ask for digitizing preview.

Approval rule: No bulk printing without a strike-off (print test) and a wash test. Otherwise you are gambling.

14) Print/Decoration Supplier Checklist

Ask these questions before you pay. If they dodge, walk away.

- What inks/films do you use (brand/type)?
- Do you print in-house or outsource?
- Can you show wash-test results or sample references?
- What is your minimum per design (and per color)?
- How do you handle misprints (reprint policy)?
- What is the timeline for strike-off approval?
- How do you ensure placement consistency (jigs/templates)?

15) Cheat Sheets

Best defaults (start here):

- T-shirts (cotton): water-based screen for premium feel; DTF for flexible small batches.
- Hoodies: embroidery or screen (bulk). Avoid huge thick transfers.
- Sportswear (poly): sublimation.
- Caps: embroidery (flat or puff).
- If you don't know your demand: POD/DTF micro-batch before bulk.

Premium perception upgrades:

- Woven labels + neck tape + clean inside finishing.
- Embroidery/chenille patches for texture.
- Better fabric weight (GSM/oz) and pre-shrink processes.
- Consistent sizing (size set approval) to reduce returns.

Final ruthless mentor line: Your brand is not your logo. Your brand is what arrives in the package. If production quality is inconsistent, marketing will not save you.